

ADAM LIM HAN JUNG

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EDUCATION

University of Sydney

M.S. (Hons) Computer Science, Advanced Entry — Data Science & AI Specialisation

Sydney, Australia

Aug. 2026 – Present

- Recipient of the Sydney International Student Award.
- Societies: SUAMS, USYD Athletics Club.

Sunway University & Lancaster University

B.S. (Hons) Computer Science

Kuala Lumpur, Malaysia

Jan. 2022 – Jan. 2025

- Coursework: Artificial Intelligence, Computer Vision, Computational Intelligence, Distributed Systems.
- Capstone: design and autonomous navigation of a quadruped robot in dynamic environments.

EXPERIENCE

Backend Java Engineer

Ant International

May 2025 – Feb. 2026

Tun Razak Exchange, Kuala Lumpur

- Integrated chargeback case-management APIs into the risk platform, improving abnormal case handling efficiency by **15%**.
- Built automated risk reporting and alerting workflows monitoring VAMP/ECP thresholds, preventing SLA breaches and enabling proactive intervention.
- Developed anomaly detection pipelines across dispute materials, chargeback patterns, and reconciliation issues, reducing escalations and supporting data-driven decisions.
- Provided on-call incident resolution and root cause analysis across cross-service failures in multi-region deployments, preventing customer impact.
- Owned end-to-end domain expertise in chargeback, pre-arbitration, arbitration, and reconciliation flows, applying event-driven logic to live case management.

Mechatronics Engineer (Software)

Logiflow Automation

Jan. 2025 – Mar. 2025

Shah Alam, Malaysia

- Integrated the Geek+ warehouse robotics API over HTTP POST/JSON to automate fleet task dispatch.
- Authored technical documentation and integration/troubleshooting guidelines adopted by the engineering team.
- Supported automation testing with data-driven optimisation of AMR routing and sensor-based workflows.

PROJECTS

Hampr — Automated Label Generation | *Client project for Ipoh Town*

Aug. 2026 – Sep. 2026

- Built a label-generation engine for a catering business (Ipoh Town), rendering print-ready labels straight from captured order data and cutting label preparation from **30 minutes to 2** per batch — a **93% reduction**.
- Replaced an Adobe Photoshop scripting dependency (ExtendScript/COM) with a pure-Python PSD rendering engine, removing the need for a licensed Photoshop install on the printing machine.
- Solved unreliable template metadata by locating text layers **positionally** rather than by name — dish name resolved via its bounding box inside the template's colour band, customer name via nearest-neighbour above it.
- Drove every dish through one generic renderer plus a per-dish resolver table instead of per-dish code, so unknown dishes skip with a warning rather than failing the whole batch.
- Bundled and pinned the required typefaces to fix silent macOS/Windows font-fallback corruption, guarded by a regression test that fails if any template font stops resolving.

Tracely — AI Observability & Root Cause Analysis Platform | *Next.js 16, React 19, Python (FastAPI), Neo4j, PostgreSQL, Inngest*

Jun. 2026 – Present

- Founded and built a platform that turns a production **traceId** into a ranked root cause *and* an executable remediation plan, replacing hours of manual on-call triage.
- Designed a multi-agent RCA engine (Planner → Investigator → Hypothesis → Synthesizer) that traces corrupt state back to its origin using **AST analysis** of the codebase, not just log correlation.
- Built a continuously reconciled knowledge graph over services, repositories, database schemas, and code entities across **Neo4j** and **PostgreSQL**, refreshed by background Inngest jobs.
- Automated remediation end to end: deterministic SQL generation for missing or corrupt data, plus a staged FixSpec → sandbox → validation pipeline that submits draft **GitHub pull requests**.

- Wired auto-triage from Slack and PagerDuty alerts with in-thread RCA summaries, ingesting traces and logs via OpenTelemetry and Grafana Loki.
- Gated every executable step behind human approval with full audit trails, so automation never mutates production state unreviewed.

PatternLoop — Personal Spaced-Repetition Study System | *TypeScript, Bun, React 19*, Jun. 2026 – Present *SQLite*

- Built a daily study system that merges algorithm practice, university coursework, and interview prep into a single “everything due today” board, replacing three separate tools and the decision of what to revise.
- Implemented a pure scheduling ladder (1 → 3 → 7 → 14 → 30 days; pass promotes, fail resets) shared verbatim by server and client so due-state can never diverge between them.
- Built an automated exam-paper pipeline that scans course material (PDFs, slides, notes), transcribes lecture recordings, and scaffolds daily practice papers per course week — with a backlog cap so missed days queue rather than dump.
- Wrote quality-control scripts detecting **answer-length tells** and **correct-option positional bias** in generated questions, wired into the test suite — turning question quality from a style guideline into a measurable regression check.
- Shipped a browser userscript that captures in-progress code straight from LeetCode’s editor, upserts the solution, auto-advances the ladder, and resets the editor for the next attempt.
- Covered the scheduler, date math, SQLite layer, and HTTP API with `bun test` against an in-memory database; mirrored the review schedule onto an embedded Google Calendar.

AI-Powered Payment & Chargeback System | *Java (Spring Boot), Python, Redis, React, Stripe* Jan. 2026 – Present

- Architected a distributed payment platform supporting top-up, refund, and chargeback workflows with strong consistency and fault tolerance across services.
- Built transaction pipelines (create intent → process payment → ledger settlement) using idempotent APIs, Redis-based distributed locks, and DB-level constraints to eliminate double-spend and race conditions.
- Propagated idempotency keys from client → backend → Stripe to guarantee exactly-once payment intent creation, with retry and DLQ mechanisms for recoverability.
- Implemented ML-driven anomaly detection for suspicious transactions, automating dispute triage and reducing manual review.
- Enforced auditability with a double-entry ledger, transactional boundaries, immutable event logs, and JWT-based stateless authentication.

Autonomous Quadruped & ROBOCON Competition Robot | *Python, ROS, Jetson Nano* Aug. 2023 – Jun. 2024

- Developed real-time event-driven navigation using ROS and depth cameras, reacting dynamically to a changing competition environment.
- Programmed sensor fusion, joint control, and autonomous pathfinding under hard real-time constraints.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, SQL, C/Arduino

Backend & Distributed: Spring Boot, FastAPI, REST/GraphQL, Microservices, Event-Driven Architecture, Message Brokers, Redis, Distributed Locks, DLQ, Idempotency

Data & AI: PostgreSQL, Neo4j, SQLite, Redis, Multi-Agent LLM Systems, AST Analysis, TensorFlow, Scikit-Learn, Computer Vision, Anomaly Detection

Web & Tooling: Next.js, React 19, Bun, Node.js, Prisma, Inngest, Flask, Docker, Git, AWS (EC2, RDS), OpenTelemetry, Grafana Loki, Datadog, ROS

Certifications: IBM Data Science Professional Certificate (Coursera, Feb. 2025)

LEADERSHIP & INVOLVEMENT

Head of Event Management

Sunway Robotics Club

Aug. 2024 – Jan. 2025

Kuala Lumpur, Malaysia

- Led the Project Developers and Event Management teams, running workshops and talks including Arduino 1.0/2.0 and 3D printing.
- Authored a reusable event-planning playbook now used by successive event managers.

Student Ambassador & Community Volunteer

Sunway University Open Day; Sunway XFarms

2023 – 2025

Kuala Lumpur, Malaysia

- Demonstrated the quadruped and combat robots to visiting students, educators, and industry professionals.
- Contributed to daily smart-farm operations (germination, planting, harvesting) alongside a volunteer team.